

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and human-in-the-loop validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures. Eliminated the analyst as the bottleneck for routine asks.
- Built and shipped to production an autonomous Jira ticket-resolution agent: 6-gate triage, MCP-driven context gathering, agentic RAG with FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months — cut routine turnaround from 30–60 minutes to under 10 minutes each. Debugged and improved it across model output, retrieval, and system logic; stakeholders never knew an agent was answering them.
- Built a hybrid orchestrator on top of the analytics agent that routes business questions through two backends in parallel — a CubeJS semantic-layer skill (co-built with the DS and product team) and the context-engineered agent — then compares outputs, reasons about disagreements, and feeds discrepancies back to refine the semantic layer's definitions over time. Evolved it into a registry-governed semantic layer where a single LLM call maps the question to strict JSON intent (the model never writes SQL) before a deterministic gate and compiler produce reproducible answers, making the one probabilistic step regression-testable.
- Built and deployed to production a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; end-to-end observability via decision logging and BigQuery state persistence, so the agent has no LLM-memory reliance. Designed a v2 architecture as a 3-tier event-driven platform self-configuring via a BigQuery event registry — new event types ship without code deploys.
- Replaced ~20 distinct recruitment workflows I had been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform; stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates, turning a recurring 3-hour task into ~3 minutes of setup.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agent & evaluation stack: MCP (Model Context Protocol) server design, sub-agent orchestration, multi-step LLM workflows, context engineering, prompt engineering, RAG (FAISS + BM25 + RRF), evaluation suites (golden sets, human-in-the-loop validation), multi-model routing across providers, headless cron agents, event-driven architectures, decision logging / observability

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

AI-assisted development: Claude Code, Cursor

APIs / backend / glue: FastAPI, Cloud Run (containerized deploy), BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

Data: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL

Viz: Streamlit, Power BI, Tableau, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL